PROBLEM 39

Particle on a String with a Plate

Original statement · Problems.pdf, p. 27

Plate Edge Contact

A heavy point particle P with mass m is connected with point O on the upper surface of a thin square plate by a massless, inextensible string of length ℓ. The distance between O and the edge of the plate is d. The string hangs over the edge of the plate and can slide without friction. You can assume that the string remains tense and in contact with the plate at all times. We denote the geometric point where the string is in contact with the plate edge as Q. We define the inertial reference frame $S_0 : \{O, \mathcal{B}_0\}$, with $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ as shown in the figure. We define the angle θ between the Ox_0 axis and the string segment OQ, and the angle ϕ between a vertical plane containing the plate edge and the line perpendicular to the plate edge that passes through P. Initially at time $t = 0, \theta(0) = \phi(0) = \pi/3$ rad, and the modulus of the velocity of P is $v_0^P(0) = \sqrt{\ell g}$, where g is the magnitude of the acceleration due to gravity.

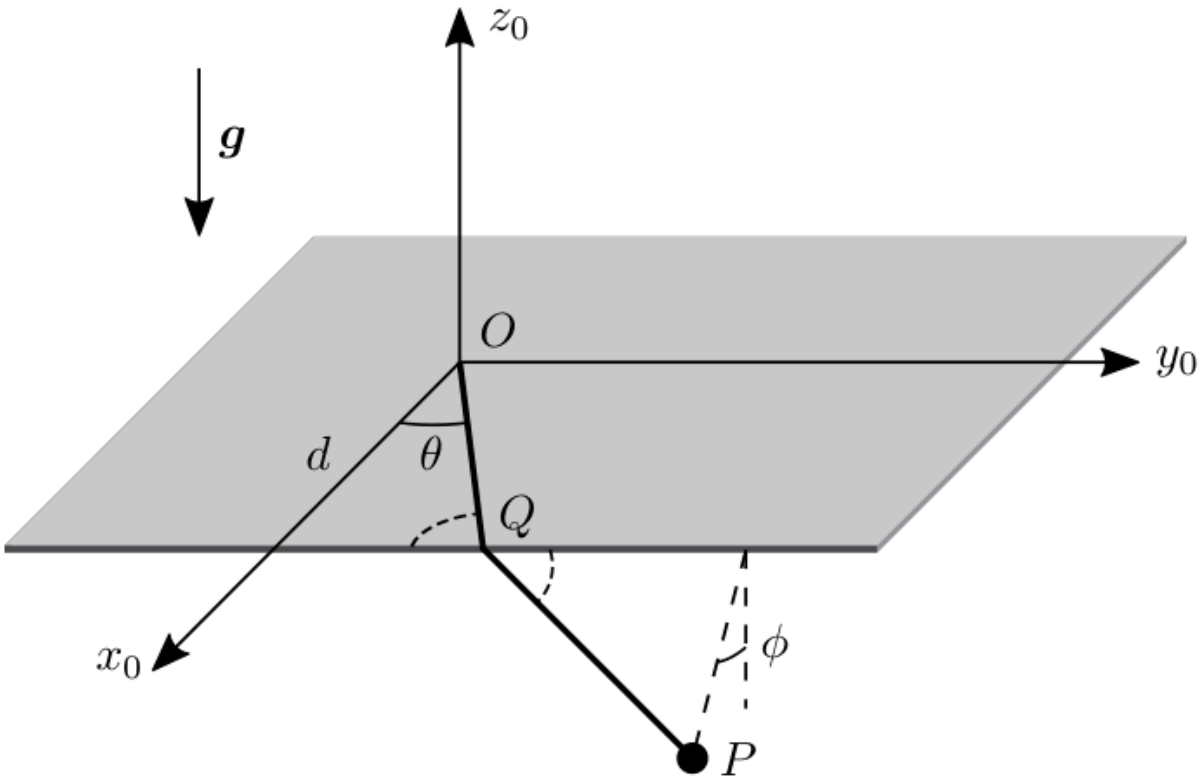


Figure: Plate with O on its upper surface, string OQ at angle θ to Ox0, hanging segment QP at angle φ, equal angles marked with dashed lines (Problems.pdf, p. 27).

(Hint: Note that under the conditions of the problem, the two angles marked with dashed lines are equal).

(a) How many degrees of freedom does the system have?

(b) Obtain the general expression of a unit vector in the direction from Q to P as a function of θ and ϕ.

(c) Obtain the general expression for the position, velocity, and acceleration vectors of Q and P in S_0 as a function of θ, ϕ, their derivatives, and the parameters of the problem.

(d) Identify and express mathematically all the forces acting on P.

(e) Find a set of second order differential equations to determine the evolution of θ and ϕ in time, free of any other unknowns.

(f) Does the tension force exert work on P? Provide proof of your answer.

(g) Derive the general expression of the mechanical energy of P in S_0 and compute its value initially. Is this a conserved quantity of motion? Justify your answer.

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a heavy particle P of mass m hangs from a massless inextensible string of length ℓ anchored at the point O of the upper surface of a thin horizontal plate. The string passes without friction over the straight edge of the plate, at perpendicular distance d from O, at the contact point Q; it remains taut and in contact with the plate.
- **Geometry (read from the figure, `Problems.pdf` PDF page 27).** The edge is the straight line $x_0 = d$, parallel to $\mathbf{j}_0$ (Ox_0 is perpendicular to the edge); θ is the angle between Ox_0 and OQ, so $\mathbf{r}_Q = d\,\mathbf{i}_0 + d\tan\theta\,\mathbf{j}_0$; and ϕ is the angle between the vertical and the line through P perpendicular to the edge, which lies in a plane normal to the edge. The statement text does not fix the direction of the edge, so this reading relies on the figure.
- **Constraints and degrees of freedom (a):** P has 3 coordinates. The string has constant length: $|OQ| + |QP| = \ell$ with Q on the edge, and Q slides on the edge (the position of Q on the edge is not fixed). The frictionless sliding over the edge makes the two string segments meet the edge at equal angles (the hint of the statement). Counting: the 3 coordinates of P minus the single length constraint give $\mathrm{CDOF} = 3 - 1 = 2$; Q adds no unknown, because the equal-angle condition fixes it once θ is known. These conditions leave the two independent coordinates θ and ϕ: $\mathrm{CDOF} = 2$.
- **Initial state:** $\theta(0) = \phi(0) = \pi/3$ and $|\mathbf{v}_P(0)| = \sqrt{\ell g}$.
- **Parameters:** m, ℓ, d, g.

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with $\mathbf{k}_0$ vertical upward; the plate in the plane $z_0 = 0$.
- **Lengths:** $|OQ| = d/\cos\theta$, so the hanging segment has length $\rho = \ell - \dfrac{d}{\cos\theta}$.
- **Swing basis.** The hanging segment QP swings in the plane normal to the edge, i.e. about the direction $\mathbf{j}_0$. The basis $\mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}_0, \mathbf{k}_1\}$ is the inertial basis rotated by $-\phi$ about $\mathbf{j}_0$ (by ϕ about $-\mathbf{j}_0$), with $\mathbf{i}_1 = \cos\phi\,\mathbf{i}_0 + \sin\phi\,\mathbf{k}_0$ and $\mathbf{k}_1 = -\sin\phi\,\mathbf{i}_0 + \cos\phi\,\mathbf{k}_0$ (columns in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\phi & 0 & -\sin\phi \\ 0 & 1 & 0 \\ \sin\phi & 0 & \cos\phi \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\phi + \sin^2\phi = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

Multiplying out, RR^T has diagonal entries $\cos^2 + \sin^2 = 1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin - \sin\cos) = 0$, so it is the identity.

Derivation

3.1 Kinematics

(b) Unit vector from Q to P

Pedagogical Justification: the edge is frictionless, so the two string segments make equal angles with it (the hint). The unit vector from Q to O is $-(\cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0)$, hence $\mathbf{e}_{QP} \cdot \mathbf{j}_0 = \sin\theta$. The remaining part of $\mathbf{e}_{QP}$ lies in the plane normal to the edge, along $\sin\phi\,\mathbf{i}_0 - \cos\phi\,\mathbf{k}_0 = -\mathbf{k}_1$, with magnitude $\sqrt{1-\sin^2\theta} = \cos\theta$:

$$\mathbf{e}_{QP} = \cos\theta\sin\phi\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0 - \cos\theta\cos\phi\,\mathbf{k}_0 = \sin\theta\,\mathbf{j}_0 - \cos\theta\,\mathbf{k}_1, \qquad |\mathbf{e}_{QP}|^2 = \cos^2\theta\left(\sin^2\phi + \cos^2\phi\right) + \sin^2\theta = 1$$

(c) Position, velocity and acceleration

Pedagogical Justification: Q is on the edge and P hangs at the distance ρ along $\mathbf{e}_{QP}$:

$$\mathbf{r}_Q = d\,\mathbf{i}_0 + d\tan\theta\,\mathbf{j}_0, \qquad \mathbf{v}_Q = \frac{d\dot\theta}{\cos^2\theta}\,\mathbf{j}_0, \qquad \mathbf{r}_P = \mathbf{r}_Q + \rho\,\mathbf{e}_{QP}$$

$$\mathbf{r}_P = \left[d + (\ell\cos\theta - d)\sin\phi\right]\mathbf{i}_0 + \left[d\tan\theta + \left(\ell - \frac{d}{\cos\theta}\right)\sin\theta\right]\mathbf{j}_0 - (\ell\cos\theta - d)\cos\phi\,\mathbf{k}_0$$

(using $\rho\cos\theta = \ell\cos\theta - d$). Write $u \equiv \ell\cos\theta - d = \rho\cos\theta \geq 0$. The $\mathbf{j}_0$ component simplifies, $d\tan\theta + \ell\sin\theta - d\tan\theta = \ell\sin\theta$, so

$$\mathbf{r}_P = (d + u\sin\phi)\,\mathbf{i}_0 + \ell\sin\theta\,\mathbf{j}_0 - u\cos\phi\,\mathbf{k}_0, \qquad z_P = -u\cos\phi, \qquad \frac{du}{d\theta} = -\ell\sin\theta$$

- **Velocity of P.** The chain rule gives $\mathbf{v}_P = \mathbf{t}_\theta\dot\theta + \mathbf{t}_\phi\dot\phi$ with the tangent vectors

$$\mathbf{t}_\theta = \frac{\partial\mathbf{r}_P}{\partial\theta} = \ell\begin{pmatrix}-\sin\theta\sin\phi\\ \cos\theta\\ \sin\theta\cos\phi\end{pmatrix}, \qquad \mathbf{t}_\phi = \frac{\partial\mathbf{r}_P}{\partial\phi} = u\begin{pmatrix}\cos\phi\\ 0\\ \sin\phi\end{pmatrix}$$

They are orthogonal ($\mathbf{t}_\theta\cdot\mathbf{t}_\phi = \ell u\sin\theta\sin\phi\cos\phi\,(-1+1) = 0$) with $|\mathbf{t}_\theta|^2 = \ell^2\left(\sin^2\theta + \cos^2\theta\right) = \ell^2$ and $|\mathbf{t}_\phi|^2 = u^2$, so

$$v_P^2 = \ell^2\dot\theta^2 + u^2\dot\phi^2$$

- **Acceleration of P.** Differentiating once more (product rule, with $\mathbf{t}_{\theta\theta} = \partial^2\mathbf{r}_P/\partial\theta^2$ and so on):

$$\mathbf{a}_P = \mathbf{t}_\theta \ddot{\theta} + \mathbf{t}_\phi \ddot{\phi} + \mathbf{t}_{\theta\theta}\dot{\theta}^2 + 2\mathbf{t}_{\theta\phi}\dot{\theta}\dot{\phi} + \mathbf{t}_{\phi\phi}\dot{\phi}^2$$

$$\mathbf{t}_{\theta\theta} = \ell \begin{pmatrix} -\cos\theta\sin\phi \\ -\sin\theta \\ \cos\theta\cos\phi \end{pmatrix}, \quad \mathbf{t}_{\theta\phi} = -\ell\sin\theta \begin{pmatrix} \cos\phi \\ 0 \\ \sin\phi \end{pmatrix}, \quad \mathbf{t}_{\phi\phi} = u \begin{pmatrix} -\sin\phi \\ 0 \\ \cos\phi \end{pmatrix}$$

- **Velocity and acceleration of Q.** From $\mathbf{v}_Q = d\dot{\theta}\cos^{-2}\theta\,\mathbf{j}_0$ and $\frac{d}{dt}\cos^{-2}\theta = 2\sin\theta\cos^{-3}\theta\,\dot{\theta}$:

$$\mathbf{a}_Q = \frac{d\left(\ddot{\theta}\cos\theta + 2\dot{\theta}^2\sin\theta\right)}{\cos^3\theta}\,\mathbf{j}_0$$

3.2 Forces

(d) Forces on P

- **Weight:** $-mg\,\mathbf{k}_0$. **String tension:** along the segment QP, toward Q: $\mathbf{T} = -T\,\mathbf{e}_{QP}$ with $T \geq 0$. The edge acts on the string, not on P. **Inertia forces:** none.

3.3 Equations of Motion

(f) Work of the tension (needed to eliminate T from the equations)

> **Pedagogical Justification:** the tension acts along $-\mathbf{e}_{QP}$ on P, so its power is $-T\,\mathbf{e}_{QP}\cdot\mathbf{v}_P$. Differentiating $\mathbf{r}_P = \mathbf{r}_Q + \rho\,\mathbf{e}_{QP}$ (product rule) gives $\mathbf{v}_P = \mathbf{v}_Q + \dot{\rho}\,\mathbf{e}_{QP} + \rho\,\dot{\mathbf{e}}_{QP}$. Since $|\mathbf{e}_{QP}| = 1$, $\mathbf{e}_{QP}\cdot\dot{\mathbf{e}}_{QP} = \frac{1}{2}\frac{d}{dt}|\mathbf{e}_{QP}|^2 = 0$, so

$$\mathbf{e}_{QP}\cdot\mathbf{v}_P = \mathbf{e}_{QP}\cdot\mathbf{v}_Q + \dot{\rho}$$

- $\mathbf{v}_Q$ has only a $\mathbf{j}_0$ component, so $\mathbf{e}_{QP}\cdot\mathbf{v}_Q = \sin\theta\,\dfrac{d\dot{\theta}}{\cos^2\theta}$. With the chain rule $\frac{d}{dt}\left(\frac{1}{\cos\theta}\right) = \frac{\sin\theta}{\cos^2\theta}\dot{\theta}$, the hanging length changes at the rate $\dot{\rho} = -\dfrac{d\dot{\theta}\sin\theta}{\cos^2\theta}$. Therefore

$$\mathbf{e}_{QP}\cdot\mathbf{v}_P=\frac{d\dot\theta\sin\theta}{\cos^2\theta}-\frac{d\dot\theta\sin\theta}{\cos^2\theta}=0\implies\mathbf{T}\cdot\mathbf{v}_P=0$$

The tension does **no work** on P ($W_T=0$).

- **Consequence.** The identity holds for arbitrary $\dot\theta$ and $\dot\phi$, so $\mathbf{e}_{QP}$ is orthogonal to both tangent vectors $\partial\mathbf{r}_P/\partial\theta$ and $\partial\mathbf{r}_P/\partial\phi$. Projecting Newton's second law on these two vectors therefore removes the tension.

(e) Second-order equations for θ and ϕ

> **Pedagogical Justification:** Newton's second law $m\mathbf{a}_P=-mg\,\mathbf{k}_0+\mathbf{T}$ is projected on $\mathbf{t}_\theta$ and $\mathbf{t}_\phi$, which are orthogonal to $\mathbf{T}$ (part f), so the tension drops out: $m\,\mathbf{a}_P\cdot\mathbf{t}_q=-mg\,\mathbf{k}_0\cdot\mathbf{t}_q$ for $q=\theta,\phi$. The scalar products of the vectors of 3.1(c) are

$$\mathbf{t}_{\theta\theta}\cdot\mathbf{t}_\theta=\ell^2\left(\cos\theta\sin\theta\sin^2\phi-\sin\theta\cos\theta+\sin\theta\cos\theta\cos^2\phi\right)=0,\qquad\mathbf{t}_{\theta\phi}\cdot\mathbf{t}_\theta=\ell^2\sin^2\theta\left(\sin\phi\cos\phi-\sin\phi\cos\phi\right)=0,\qquad\mathbf{t}_{\phi\phi}\cdot\mathbf{t}_\theta=\ell u\sin\theta$$

$$\mathbf{t}_{\theta\theta}\cdot\mathbf{t}_\phi=0,\qquad\mathbf{t}_{\theta\phi}\cdot\mathbf{t}_\phi=-\ell u\sin\theta,\qquad\mathbf{t}_{\phi\phi}\cdot\mathbf{t}_\phi=0$$

and the orthogonality of $\mathbf{t}_\theta$ and $\mathbf{t}_\phi$ gives

$$\mathbf{a}_P\cdot\mathbf{t}_\theta=\ell^2\ddot\theta+\ell u\sin\theta\,\dot\phi^2,\qquad\mathbf{a}_P\cdot\mathbf{t}_\phi=u^2\ddot\phi-2\ell u\sin\theta\,\dot\theta\dot\phi$$

while $-g\,\mathbf{k}_0\cdot\mathbf{t}_\theta=-g\ell\sin\theta\cos\phi$ and $-g\,\mathbf{k}_0\cdot\mathbf{t}_\phi=-gu\sin\phi$. Dividing the first projection by ℓ and the second by $u>0$:

$$\boxed{\ell\ddot\theta+(\ell\cos\theta-d)\sin\theta\,\dot\phi^2+g\sin\theta\cos\phi=0}$$

$$\boxed{(\ell\cos\theta - d)\,\ddot{\phi} - 2\ell\sin\theta\,\dot{\theta}\dot{\phi} + g\sin\phi = 0}$$

- **Tension (afterwards).** Along $\mathbf{e}_{QP}$: $T = mg\cos\theta\cos\phi + m\,\mathbf{v}_P\cdot\dot{\mathbf{e}}_{QP}$, since $\mathbf{a}_P\cdot\mathbf{e}_{QP} = -\mathbf{v}_P\cdot\dot{\mathbf{e}}_{QP}$ (from $\mathbf{v}_P\cdot\mathbf{e}_{QP} = 0$).

3.4 Integration and Answers

(g) Mechanical energy

> **Pedagogical Justification:** the plate is at rest (scleronomic constraint), the tension does no work (part f) and gravity is conservative, so the mechanical energy of P is conserved:

$$E = \tfrac{1}{2}mv_P^2 + mgz_P = \tfrac{1}{2}m\left[\ell^2\dot{\theta}^2 + (\ell\cos\theta - d)^2\,\dot{\phi}^2\right] - mg\,(\ell\cos\theta - d)\cos\phi = \text{const}$$

- **Initial value.** At $t = 0$, $|\mathbf{v}_P|^2 = \ell g$ and $\cos\theta = \cos\phi = \tfrac{1}{2}$, so $z_P(0) = -\left(\tfrac{\ell}{2} - d\right)\tfrac{1}{2}$:

$$\boxed{E_0 = \tfrac{1}{2}m\ell g - mg\left(\frac{\ell}{2} - d\right)\frac{1}{2} = \frac{mg\ell}{4} + \frac{mgd}{2}}$$

It is a conserved quantity of the motion.

Interpretation & Checks

- **Variable-length spherical pendulum:** the system acts as a spherical pendulum whose length changes as Q slides along the edge.
- **Limit $d \to 0$:** the edge passes through O, so $Q = O$, $\rho = \ell$ and $\mathbf{e}_{QP}$ is the unit vector of a spherical pendulum of length ℓ with $\cos\gamma = \cos\theta\cos\phi$ the cosine of its angle with the downward vertical; the energy reduces to $E_0 = mg\ell/4$, equal to $\tfrac{1}{2}m\ell g - mg\ell\cos\theta\cos\phi$ at $\theta = \phi = \pi/3$.
- **Dimensional check:** $[\rho] = \mathrm{m}$; $[mg\ell] = [mgd] = \mathrm{J}$; $[u] = \mathrm{m}$ and every term of the two equations of motion has units $\mathrm{m/s^2}$.

- **Check against the official key:** the key of `Problems.pdf` answers (f): "No, the tension does not exert work; it is easy to prove it with $\mathbf{e}_r \cdot \mathbf{v}_P^0 = 0$", which is the result of 3.3. The rest is verified by independent derivation. Sympy confirms $\mathbf{e}_{QP} \cdot \mathbf{v}_P = 0$, $v_P^2 = \ell^2\dot\theta^2 + (\ell\cos\theta - d)^2\dot\phi^2$, $z_P = (d - \ell\cos\theta)\cos\phi$, the equivalence of the two projections with the equations above, and numerically $E_0 = mg\ell/4 + mgd/2 = 7.3575$ J for $m = 1$ kg, $\ell = 2$ m, $d = 0.5$ m, equal to $\tfrac{1}{2}m\ell g + mgz_P(0) = 9.81 - 2.4525$ J.

Source: Problems.pdf